Russell Day

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Education

M.S. Computational Medicine and Bioinformatics — Ann Arbor, MI
DCMB, Michigan Medicine — University of Michigan — Accelerated Master's Program — *Aug. 2026 – Present*

B.S. Data Science; Minor in Business — Ann Arbor, MI
University of Michigan — College of Literature, Science, and the Arts (LSA), GPA: 3.869/4.000 — *Aug. 2023 – Present*

- Relevant Coursework: EECS280, EECS281, EECS445, EECS484, EECS485, MATH217, STATS425, STATS426; Minor in Business through the Ross School of Business (Aug. 2025 – Present)

Research Experience

Research Assistant — Dec. 2024 – Present
CAD-AI Lab · Michigan Medicine Radiology — *Ann Arbor, MI*

- Finetuned a 3B-parameter LLM (96% accuracy) to mirror GPT-4 performance (94%) for targeted clinical information extraction under HIPAA constraints
- Combining multimodal data (images + text) to model five-year survival likelihood for bladder cancer patients

Research Assistant Volunteer — Sep. 2021 – Present
Department of Orthopedics and Neurosurgery · Henry Ford Health — *Detroit, MI*

- Conducted research on gender and racial diversity in orthopedic medicine, contributing to three peer-reviewed publications
- Implemented internal LLM pipelines for systematic literature reviews, reducing review time by 60%

Research Assistant — Sep. 2023 – Dec. 2024
Aton Lab · University of Michigan (Media) — *Ann Arbor, MI*

- Topic: Effect of sleep on memory consolidation
- Working in a wet lab with mice, collecting and analyzing data

Work Experience

SDE Intern — May 2026 – Jul. 2026
Amazon · Alexa AI — *Boston, MA*

- Built full-stack pipeline (React, Python/FastAPI) to streamline evaluation metric onboarding on LLM-based agentic AI benchmarking platform
- Reengineered backend from SQL to JSON config, reducing required code $\sim$10$\times$ (133 $\rightarrow$ 11 lines) and turnaround from 3–4 weeks to 1–2 days
- Built on AWS (CDK, Lambda, API Gateway, DynamoDB, SNS, IAM); enabled faster iteration for Alexa AI evaluation workflows

ML Intern — Jan. 2025 – Dec. 2025
Trinity Health — *Ann Arbor, MI*

- Develop ML models to audit billing compliance across >$20M ARR tied to primary care revenue streams
- Implement generative AI pipelines that process millions of patient notes to surface miscoding risk and automate coding reviews

AI Automation Engineering Intern — May 2024 – Aug. 2024
Protera Health — *Troy, MI*

- Built Python applications that paired Generative AI with Google Search to automate high-precision data collection
- Processed 1k+ records through custom ETL pipelines for lead analysis and categorization to inform outreach

Technical Skills

Languages: Python, R, SQL, JavaScript/TypeScript, HTML/CSS, VBA, C++
Tools & Frameworks: React, FastAPI, AWS (CDK/Lambda/API Gateway/DynamoDB/SNS/IAM), Streamlit, LangChain, Pydantic, Pandas, PyTorch
Specializations: Research, LLMs, Computer Vision, NLP, Clinical Data Analysis, Machine Learning, AI-Agent Coding

Publications & Presentations

Publications

- Cross, A. G., Rain, U. Z., Makhni, E. C., Schnettler, E. N., Jayakumar, P., Berry, S., Vandenboom, J., Keteyian, C., **Day, R.**, Verhagen, D., Schumacher, M., Kucksdorf, J., & Muma, B. (2025). Clinical and engagement efficacy of a virtual musculoskeletal integrated practice unit in health system employees: aggregate and comparative results from two employers. *Frontiers in Rehabilitation Sciences*, 6, 1541508.
- Meadows, A. M., Skinner, M. M., Hazime, A. A., **Day, R. G.**, Fore, J. A., & Day, C. S. (2023). Racial, Ethnic, and Sex Diversity in Academic Medical Leadership. *JAMA Network Open*, 6(9), e2335529.
- Meadows, A. M., Skinner, M. M., Faraj, M. T., Hazime, A. A., **Day, R. G.**, Fore, J. A., & Day, C. S. (2022). Racial, Ethnic, and Gender Diversity in Academic Orthopaedic Surgery Leadership. *JBJS*, 104(13), 1157-1165.

Conference Proceedings

- **Day, R.**, Hadjiiski, L., Sun, D., Chan, H. P., Caoili, E., Cohan, R. H., Alva, A., Bruno, G., Zhou, C., & Gulani, V. (2026). LoRA Fine-Tuning of a Small 3B LLM for Clinical Information Extraction and Five-Year Survival Prediction in Bladder Cancer. *SPIE Medical Imaging 2026: Imaging Informatics*, Proc. SPIE 13930, Vancouver, Canada. [Oral]
- Yordanova, A., Hadjiiski, L., **Day, R.**, Fiori, S., Caoili, E. M., Cohan, R. H., Chan, H. P., Worden, F., Hammer, G., Wilms, M., & Zhou, C. (2026). Integrated System for Characterization of Malignant and Benign Adrenal Lesions and Differentiation of High- and Low-Risk Adrenal Cancer on CT Scans Using Radiomics-Based Decision Support. *SPIE Medical Imaging 2026: Computer-Aided Diagnosis*, Proc. SPIE 13926, Vancouver, Canada.

Podium Presentations

- **Day, R.**, Hadjiiski, L., Sun, D., Chan, H. P., Caoili, E., Cohan, R. H., Bruno, G., Zhou, C., & Gulani, V. (2027). LoRA Fine-Tuning of a Compact 3B LLM for Clinical Information Extraction from Complete Perioperative Records and Five-Year Survival Prediction in Bladder Cancer. *SPIE Medical Imaging 2027: Imaging Informatics*, Vancouver, Canada, Feb. 16, 2027. [Accepted, Oral]
- **Day, R. G.**, Hadjiiski, L. M., Sun, D., Chan, H. P., Caoili, E. M., Cohan, R. H., Alva, A. S., Bruno, G., Zhou, C., Gulani, V. (2025). Performance Evaluation of a Fine Tuned Smaller Large Language Model (LLM) for Extracting Clinical Predictors of 5 Year Survival in Bladder Cancer Patients Post Cystectomy. *RSNA Annual Conference*, Chicago, IL. [Podium]

Invited Talks

- Invited Speaker. *Large Language Models in Clinical Workflows.* Michigan Institute for Imaging Technology Translation (MIITT), Ann Arbor, MI (November 2025).

Poster Presentations

- **Day, R. G.**, Hadjiiski, L. M., Sun, D., Chan, H. P., Caoili, E. M., Cohan, R. H., Wilms, M., Bruno, G., Zhou, C., & Gulani, V. (2026). LoRA versus Full Fine-Tuning of a Local Lightweight LLM for Extracting 5-Year Survival Predictors from Unstructured Reports after Radical Cystectomy for Bladder Cancer. *RSNA 112th Scientific Assembly and Annual Meeting*, Chicago, IL, Nov. 29–Dec. 3, 2026. [Poster]
- Fiori, S., Hadjiiski, L., Yordanova, A., Caoili, E. M., Cohan, R. H., Chan, H. P., **Day, R.**, Worden, F., Hammer, G., & Zhou, C. (2025). Computerized Decision Support for High- and Low-Risk Adrenal Cancer Diagnosis in CT Using Radiomic Biomarkers. *RSNA Annual Meeting*, Chicago, IL. [Poster]
- **Day, R.G.**, Dunaway, C., Chen, Z., Gudi, M., Day, C.S. (2025). Leveraging Artificial Intelligence Models to Streamline Systematic Reviews in Hand Surgery. *AAHS Annual Conference*, Waikoloa, HI. [Poster]
- McDonald, K.O., Prabhu, A.S., **Day, R.G.**, Aton, S.J. (2024). Chemogenetic Silencing of Dentate Gyrus Granule Cells During Memory Consolidation May Disrupt Contextual Fear Memory. *MIP Annual Research Symposium*, Ann Arbor, MI.

Panels & Moderation

- Panelist, "Artificial Intelligence in Orthopaedic Surgery Education," AOA/CORD Summer Conference; 2025 June 21; St. Louis, MO, USA

Student Organizations & Projects

President Apr. 2026 – Present

Michigan Data Science Team *Ann Arbor, MI*

- Leading organization-wide strategy across onboarding, education, and project initiatives

Vice President of Education *Apr. 2025 – Apr. 2026*

- Leading the education committee which supervises all of MDST's onboarding and educational initiatives
- Organize and lead workshops covering Python fundamentals, Git workflows, and data visualization best practices

Project Lead, Breast Cancer Classification using CV (Link) *Dec. 2024 – Apr. 2025*

- Led a team of 10+ students to develop vision-based ML pipelines on the BreakHis dataset
- Trained ViT models with subtype-specific objectives (AUC = 0.989), improving BoW baselines by +23.7% and CNNs by +11.2%

Consulting Data Analyst for Michigan Medicine *Aug. 2024 – Dec. 2024*

- Worked with the UM School of Medicine to clean, analyze, and pipeline faculty career data using Python/Pandas, SQL, and Tableau

Vice President of Technology Dec. 2024 – Present

Blueprints For Pangaea *Ann Arbor, MI*

- Leading the inaugural technology team to define product vision and internal tooling across 12 chapters nationwide
- Creating SOPs to define and structure cross-chapter operations and tech handoffs
- Coordinating internal tooling rollouts targeting a 50% productivity lift for logistics and inventory teams

Project Lead, Inventory Automation (B4P) *May 2024 – Apr. 2025*

- Improved inventory efficiency by 50% by leading a team of five to deploy an automation pipeline processing 10,000+ SKUs
- Integrated tracking workflows to monitor more than $100k of medical supplies annually across global relief shipments

Intern *Jan. 2024 – May 2024*

- Facilitated partnership between the club and a non-profit organization for a shipment of medical supplies
- Helped develop a comprehensive guide to better their understanding of the club's mission and operations
- Cleaned and organized data to improve inventory efficiency for the app design team

Volunteer Experience

Volunteer Jun. 2025 – Present

Food Gatherers Community Kitchen *Ann Arbor, MI*

- Cooking, serving, and cleaning up hot meals for guests at the Robert J. Delonis Center

Volunteer Jan. 2025 – Present

Crisis Text Line *Ann Arbor, MI*

- Providing crisis support and mental health resources to individuals in need

Volunteer Jan. 2024 – Present

Kidney Disease Screening and Awareness Program *Ann Arbor, MI*

- Participating in community health screening initiatives

Volunteer July 2024, June 2022, March 2018

Breanna's House of Joy *Chiang Mai, Thailand*

- Volunteered at children's home providing educational and recreational support

Honors & Awards

University Honors for all eligible Semesters (2023-2025), James B. Angell Scholar 3/16/2025